Flow is All You Need for WiFi: Rectifying 3D Human Poses without Complex Architectures

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Abstract

WiFi-based human pose estimation offers an occlusion-resilient alternative to camera-based sensing, but estimating multi-person 3D skeletons from channel state information remains difficult because WiFi signals are noisy, low-dimensional, and strongly affected by multipath propagation. This paper studies a compact draft-to-refine redesign of the Person-in-WiFi 3D pipeline. Starting from the PETR-style baseline, we introduce motion-aware spectral tokenization, a flattened Mamba2-CSI encoder, a lightweight WiTiDAR pose head, and two-step flow-based refinement. Under the internal 20-epoch fixed-layout ablation protocol, the selected M9_RF2 configuration achieves 165.487 mm MPJPE, improving over the re-evaluated M0 baseline by 7.067 mm while increasing throughput from 118.7 FPS to 209.9 FPS, reducing parameters from 13.130M to 3.618M, and reducing peak memory from 155.60 MB to 25.29 MB. The title is read as a scoped architectural statement about replacing a complex pose-decoding pipeline with compact sequence modeling and lightweight flow refinement, not as an unrestricted claim about WiFi sensing or flow in isolation.

Index Terms

WiFi sensing, CSI, 3D human pose estimation, Mamba2, flow matching, spectral tokenization.

I. Introduction

WiFi channel state information (CSI) provides a non-visual sensing signal that can be used for human perception in indoor environments. Unlike camera-based pose estimation, WiFi sensing can operate under occlusion and low-light conditions, but the input is indirect: body motion perturbs multipath propagation rather than producing explicit visual evidence. This makes multi-person 3D pose estimation from WiFi substantially harder than pose estimation from images.

Person-in-WiFi 3D demonstrated an end-to-end PETR-style formulation for multi-person 3D pose estimation from WiFi CSI [1]. In this work, we ask whether that pipeline can be made smaller and faster while preserving or improving pose accuracy. Our redesign follows the full M0–M9 development path from the original baseline to a compact draft-to-refine model. The selected M9_RF2 configuration combines spectral tokenization, flattened Mamba2-CSI sequence modeling, a WiTiDAR draft head, and two-step flow refinement.

This study is scoped as an application-oriented fixed-layout ablation on the Person-in-WiFi 3D-style sensing setup, with the goal of identifying a compact operating point for the three-receiver ResFes prototype rather than making an unrestricted claim across WiFi sensing environments.

The phrase "Flow is All You Need" is therefore intended as a scoped architectural claim. In the fixed-layout Person-in-WiFi 3D setting studied here, lightweight flow refinement serves as the final correction stage after a compact backbone, allowing the selected model to avoid returning to a complex Transformer-style pose decoder. This interpretation is reflected in the subtitle: rectifying 3D human poses without complex architectures.

The contribution is an internally controlled ablation rather than a new benchmark claim. The study traces spectral tokenization, PETR and WiTiDAR heads, Transformer and Mamba-family encoders, draft-only variants, and one- versus two-step flow refinement across M0–M9. We compare MPJPE, scene cardinality, throughput, parameter count, and peak memory under the same fixed-layout evaluation, then report flow, per-joint, and receiver-subset diagnostics. The evidence identifies M9_RF2 as the strongest observed accuracy-efficiency operating point among the tested full-paper configurations.

II. Related Work

A. Human Pose from Images

Camera-based pose estimation is the most mature paradigm and supplies the conceptual templates this paper inherits [2]. Bottom-up methods such as OpenPose [3] regress per-pixel keypoint heatmaps and part affinity fields and then associate them into instances, which scales gracefully with crowd size. Top-down methods such as AlphaPose [4] first run a person detector and then a single-person estimator, usually attaining higher accuracy at the cost of cropping overhead. High-resolution backbones such as HRNet [5] sharpen the accuracy ceiling by preserving spatial resolution across the network. These methods motivate the set-prediction perspective we later adopt for WiFi but inherit the deployment limits of cameras—appearance capture, lighting dependence, and reduced reliability under occlusion—which conflict with privacy-sensitive indoor monitoring.

B. Human Pose from Wi-Fi

Radio and WiFi sensing developed from through-wall pose recovery and ambient physiological or gesture sensing [6]–[9]. CSI-based representations then supported person perception and pose recovery without an image stream [10]–[13]. WiNect and GoPose advanced free-form single-person 3D tracking [14], [15], while MM-Fi broadened multimodal wireless benchmarks [16]. Application-oriented systems have also considered metaverse avatars, joint activity–pose analysis, and industrial skeleton monitoring [17]–[19]. Recent graph-based and lightweight continuous-pose studies further illustrate the range of structural and efficiency choices in WiFi pose modeling [20], [21].

Person-in-WiFi 3D adapts the detection-and-regression perspective to WiFi CSI and provides the baseline used in this study [1]. A related short-form study also reports 3D WiFi pose estimation under commodity sensing [22]. Transformer architectures became important for structured prediction after the success of attention-based sequence modeling [23], while the DETR family of set predictors [24] and its pose-estimation extension PETR [25] provide the architectural template we modify.

State space models such as Mamba provide linear-time sequence modeling with selective recurrence [26]. Vision Mamba, VMamba, and LocalMamba show how selective state-space computation can be adapted to structured visual tokens [27]–[29]. Mamba-2 further connects structured state space models and attention-like computation [30]. These properties motivate replacing a heavier Transformer encoder with a compact Mamba2-CSI encoder for WiFi token sequences.

Rectified flow learns a velocity field for transporting samples between source and target distributions [31]. Diffusion-based 3D pose methods have likewise used iterative generation, multi-hypothesis aggregation, and learned refinement to improve pose structure [32]–[34]. Here, however, the source is not random noise; it is a draft 3D pose produced by the detection head. The flow module is therefore used as a learned pose refinement operator. Because no reflow stage is performed, the interpretation is closer to conditional flow-matching refinement than a full generative rectified-flow pipeline.

Recent WiFi-pose work separates fixed-benchmark pose estimation from robustness-focused protocols: Person-in-WiFi 3D uses a fixed receiver layout with receiver-count analysis, while DT-Pose studies domain-consistent representation learning as a dedicated cross-domain problem [35]. We follow the former convention to focus on compact architecture design and application-scale inference.

III. Wi-Fi Signals

A. Channel State Information

Commodity WiFi devices communicate over N orthogonal subcarriers; for each subcarrier i the received and transmitted symbols are related by

$$Y_i = H_i X_i + n_i, \quad i \in [1, N], \tag{1}$$

where $H_i \in \mathbb{C}$ is the channel state information (CSI), X_i the transmitted symbol, and n_i additive noise. Each H_i encodes the joint effect of path loss, scattering, reflection, and small-scale fading along the propagation path, so its amplitude and phase vary with the body geometry, motion, and pose of people between transmitter and receiver. This dependency makes WiFi a viable sensing modality for pose, but it also means the signal aggregates contributions from the whole scene rather than localizing them to a specific person, which is fundamentally harder than visual input.

B. Phase Sanitization

Raw CSI phase is corrupted by carrier-frequency offset, sampling-frequency offset, and packet detection delay, which together appear as a per-packet linear drift across subcarriers. Standard sanitization techniques fit and remove this drift before the signal is consumed by a learning model [36]. We follow the project default and apply discrete-wavelet amplitude denoising together with linear-phase normalization at the dataset stage; this leaves both amplitude and a stabilized phase available to the downstream pipeline. The denoising stage is treated as a fixed preprocessing component and is not retrained.

C. Hardware Configuration

The sensing layout uses one transmitter and three receivers around a 3.5×4 m indoor area. Each receiver contributes three antenna streams, and each stream provides 30 active subcarriers over a 20-packet temporal window. After amplitude and phase are concatenated, the transmitter dimension is omitted because only one transmitter is used, leaving a $3 \times 3 \times 20 \times 60$ tensor that is flattened into the 180 WiFi tokens consumed by the encoder in Section IV. Section V-A gives the recording protocol.

IV. Method

A. Overview

Figure 1 summarizes the pipeline. A preprocessed CSI tensor is converted into WiFi tokens, encoded by a compact sequence model, decoded into person hypotheses, and refined from draft pose to final pose through a flow-based correction module.

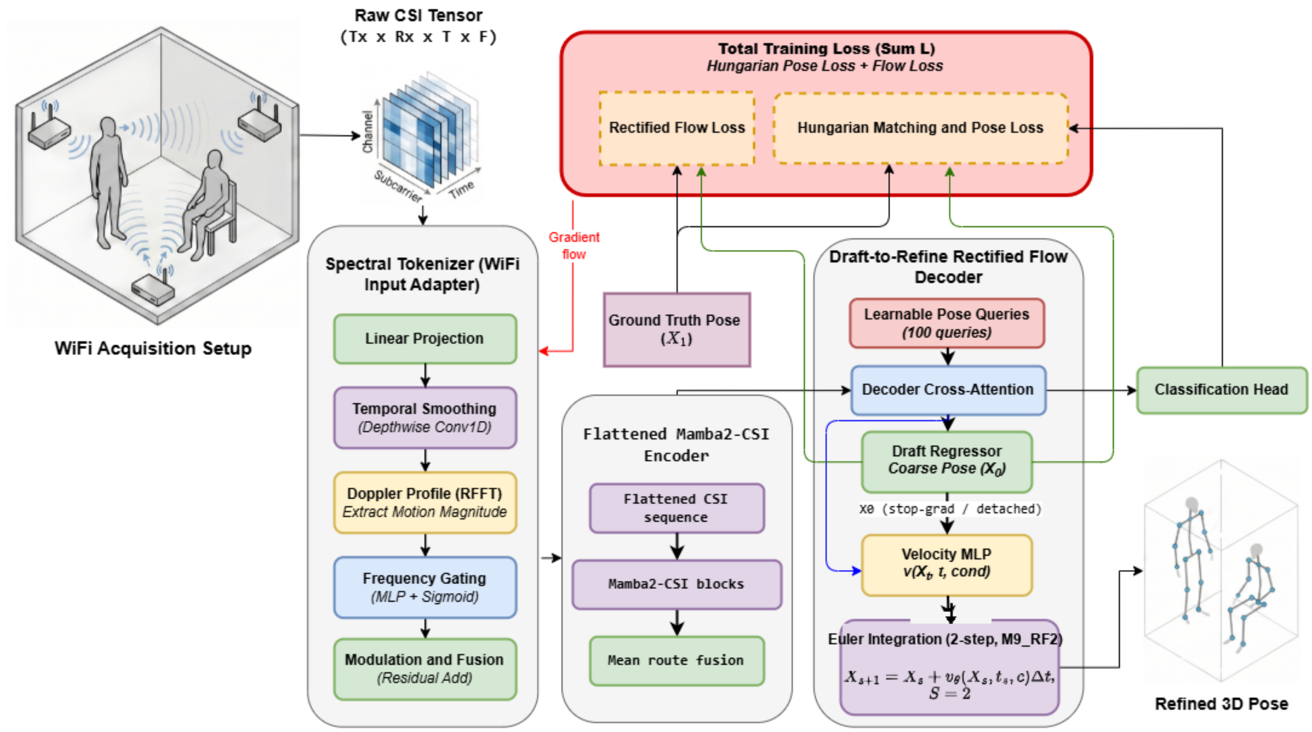

Fig. 1. System overview of the WiTiDAR draft-to-refine pipeline.

B. Motion-Aware Spectral Tokenization

The baseline uses a linear projection from CSI features to token embeddings. The spectral tokenizer keeps this linear path as a residual bypass and adds a motion-aware temporal branch. The projected sequence is reshaped into spatial groups and temporal steps. A depthwise temporal convolution smooths local temporal variation, and a real FFT over time estimates a spectral motion profile. A lightweight frequency-gating module maps this profile back to time-step weights.

C. Mamba2-CSI Encoder

The Mamba2-CSI encoder replaces a Transformer-heavy encoder with a compact sequence model. In the selected flattened configuration, tokens are processed as a single WiFi sequence rather than through multiple manually routed branches. This reduces parameter count and latency while retaining sequence context across spatial and temporal CSI tokens.

D. WiTiDAR Draft Head and Flow Refinement

The WiTiDAR head predicts person hypotheses through learned pose queries. During training, Hungarian assignment and the classification, keypoint, and bone losses are computed on draft predictions. For matched positives, the flow module receives the detached draft pose, the target pose, and the query condition. The training objective follows a conditional velocity-matching form:

$$x_t = (1-t)x_0 + tx_1, \tag{2}$$

$$v^*(x_t, t, c) = x_1 - x_0, \tag{3}$$

$$\mathcal{L}_{flow} = \|v_\theta(x_t, t, c) - v^*(x_t, t, c)\|_2^2. \tag{4}$$

At inference, Euler integration updates the pose:

$$x_{k+1} = x_k + \Delta t\, v_\theta(x_k, t_k, c). \tag{5}$$

M9 uses one Euler step, while M9_RF2 uses two steps on the same trained checkpoint before reporting the final 3D keypoints.

E. Implementation Details

The selected M9_RF2 model uses 100 learned person queries, 256-dimensional embeddings, and 14 output keypoints. CSI is first mapped by a spectral `WifiInputAdapter`; the flattened Mamba2-CSI encoder has three layers with d_{state}=64, d_{conv}=4, expansion 2, head dimension 64, dropout 0.1, a `time_major` route, mean route fusion, no positional embedding, and no final attention block. The WiTiDAR head applies 8-head query attention, a two-layer MLP draft regressor, and a binary classification head. The flow refiner is a conditional `VelocityMLP` over 3D keypoint coordinates and query-level features.

Training uses focal classification loss with weight 2.0, keypoint L1 loss with weight 5.0, bone-length loss with weight 2.0, and flow loss with weight 10.0 for the rectified-flow variants. All reported models use AdamW with learning rate 2×10^{-5} and weight decay 0.05. At evaluation time, predictions are assigned to ground truth with Hungarian matching under a 500 mm threshold. Unmatched ground-truth persons receive a 500 mm penalty in overall MPJPE; Matched MPJPE excludes this penalty and averages only over assigned prediction–ground-truth pairs.

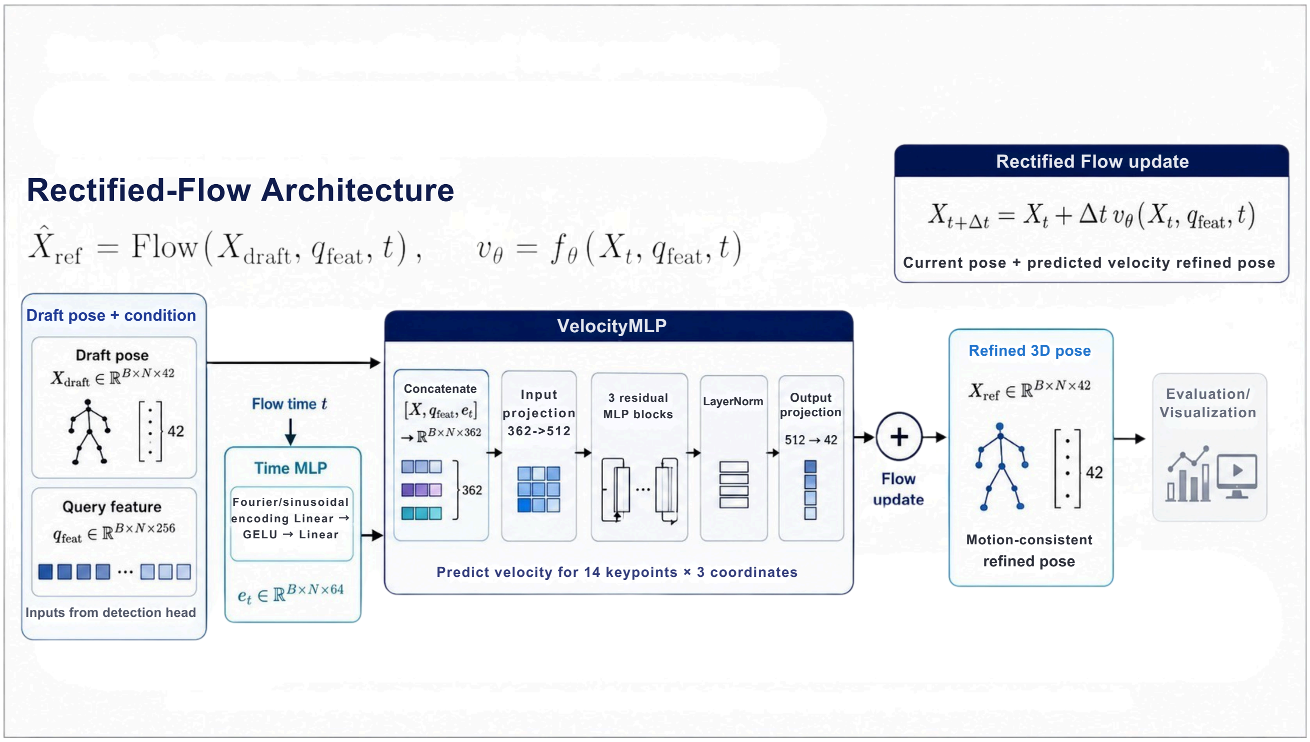

Fig. 2. Conditional pose-flow refinement from draft coordinates to refined 3D keypoints.

V. Experimental Protocol

A. Dataset and Acquisition

Data were recorded with four ThinkPad X201 laptops equipped with Intel 5300 WiFi adapters, following the Person-in-WiFi 3D acquisition protocol [1]. One laptop transmitted on channel 128 at 5.64 GHz through a single antenna, while the other three laptops recorded CSI through three antennas each. The transmitter generated 300 packets per second, with 30 subcarriers retained for every transmitter–receiver link. An Azure Kinect simultaneously captured RGB-D observations at 15 frames per second. The camera and the three receiving streams were aligned before each recording session, so one pose annotation could be paired with a CSI block of size $1 \times 3 \times 3 \times 30 \times 20$, corresponding to transmitter, receiver, antenna, subcarrier, and time dimensions. Concatenating amplitude with sanitized phase changes the last feature dimension from 30 to 60; removing the singleton transmitter axis and reordering the remaining axes produces the $3 \times 3 \times 20 \times 60$ model input and its 180 tokens.

Seven volunteers (heights 160–177 cm, weights 55–70 kg) performed eight daily actions (reaching out, raising hands, bending over, stretching, sitting down, lifting legs, standing, walking) across three locations (office, classroom, corridor). The resulting dataset contains 89,946 training and 7,824 test samples partitioned by scene cardinality (1P / 2P / 3P), summarized in Table I.

B. Training Setup

All configurations were trained for 20 epochs with the AdamW optimizer at a base learning rate of 2×10^{-5}, weight decay 0.05, batch size 2, and FP32 precision on a single CUDA GPU. Each configuration uses a fixed random seed for

TABLE I
DATASET STATISTICS FOR THE PERSON-IN-WIFI 3D INTERNAL TEST BED USED THROUGHOUT THIS PAPER. COUNTS ARE CSI SAMPLES; EACH SAMPLE IS PAIRED WITH A SYNCHRONIZED AZURE KINECT SKELETON.

split	1-person	2-person	3-person	total
training	28,121	36,242	25,583	89,946
test	2,586	3,184	2,054	7,824
total	30,707	39,426	27,637	97,770

reproducibility within this case study. CSI inputs use the project preprocessing pipeline (discrete-wavelet amplitude denoising and phase normalization). Latency was measured with batch size 1 in FP32 after 20 warm-up passes and 100 measured passes with `torch.cuda.synchronize()` between iterations; reported latency is the mean and throughput is its reciprocal. This protocol keeps the comparison focused on application-scale architecture choices under the same training and deployment budget.

C. Evaluation Metrics

For a predicted 3D pose $\hat{P}$ and the matched ground truth P_{gt}, the per-joint position error (PJPE) of joint k is the Euclidean distance between predicted and ground-truth coordinates, and the mean per-joint position error (MPJPE) averages PJPE over the $K{=}14$ keypoints [37]:

$$\mathrm{PJPE}(k) = \left\|\hat{P}_k - P_{gt,k}\right\|_2, \quad \mathrm{MPJPE} = \frac{1}{K}\sum_{k=1}^{K}\mathrm{PJPE}(k). \tag{6}$$

To diagnose where error concentrates, we additionally report the per-joint dimension localization error (PJDLE), defined as the ℓ_1 distance along each spatial axis $a \in \{h, v, d\}$:

$$\mathrm{PJDLE}_a(k) = \left|\hat{P}_{k,a} - P_{gt,k,a}\right|, \quad \mathrm{MPJDLE}_a = \frac{1}{K}\sum_{k=1}^{K}\mathrm{PJDLE}_a(k). \tag{7}$$

Per-frame errors are aggregated over the fixed test protocol, separated by scene cardinality (1P/2P/3P), and reported jointly as the overall MPJPE for ranking. The supplementary flow analysis also reports Matched MPJPE and the unmatched ground-truth person count.

VI. RESULTS

A. Main Ablation

Table II defines the full M0–M9 development path and the selected two-step M9_RF2 evaluation. Table III reports accuracy and efficiency from the full-paper experiment logs. M0–M3 retain PETRHead while varying the input and encoder; M4–M6 introduce the lightweight WiTiDAR draft head without flow; M7–M9 add rectified-flow refinement under Transformer, Mamba-1, and flattened Mamba2-CSI encoders.

TABLE II
MODEL VARIANT SUMMARY FOR THE FULL-PAPER ABLATION PATH.

ID	Input	Encoder	Pose Head	Flow	Role in Study
M0	Linear	Transformer	PETRHead	No	Original baseline
M1	Spectral	Transformer	PETRHead	No	Tokenizer ablation
M2	Spectral	Mamba-1	PETRHead	No	Encoder ablation
M3	Spectral	Mamba2-CSI	PETRHead	No	Encoder ablation
M4	Spectral	Transformer	WiTiDAR draft	No	Draft-head ablation
M5	Spectral	Mamba-1	WiTiDAR draft	No	Draft-head ablation
M6	Spectral	Mamba2-CSI	WiTiDAR draft	No	Compact draft model
M7	Spectral	Transformer	WiTiDAR	1-step	Flow model
M8	Spectral	Mamba-1	WiTiDAR	1-step	Flow model
M9	Spectral	Mamba2-CSI	WiTiDAR	1-step	Compact flow model
M9_RF2	Spectral	Mamba2-CSI	WiTiDAR	2-step	Selected evaluation

TABLE III
FULL 20-EPOCH ABLATION. LOWER MPJPE, PARAMETER COUNT, AND MEMORY ARE BETTER; HIGHER FPS IS BETTER.

ID	MPJPE	1P	2P	3P	FPS	Params (M)	Memory (MB)
M0	172.554	136.141	168.012	192.529	118.7	13.130	155.60
M1	169.271	132.597	163.323	190.810	83.1	13.199	155.86
M2	174.961	143.475	166.642	196.772	69.4	11.967	151.16
M3	175.658	140.116	169.358	197.084	102.0	10.292	142.69
M4	169.796	126.479	167.593	190.251	180.8	5.255	31.61
M5	171.055	127.258	171.979	188.481	116.8	4.023	26.80
M6	167.573	125.532	164.528	**188.363**	**247.7**	**1.813**	**18.41**
M7	168.342	127.129	162.775	191.390	145.8	7.059	38.49
M8	169.653	122.253	165.397	193.943	109.4	5.827	33.68
M9	168.086	125.958	161.832	192.229	206.9	3.618	25.29
M9_RF2	**165.487**	**121.785**	**159.341**	190.179	209.9	3.618	25.29

M9_RF2 improves MPJPE by 7.067 mm over the re-evaluated M0 baseline and by 2.599 mm over one-step M9. It reaches 209.9 FPS with 3.618M parameters and 25.29 MB peak memory, compared with 118.7 FPS, 13.130M parameters, and 155.60 MB for M0. M6 remains the smallest and fastest draft-only model, while M9_RF2 provides the best observed complete-model accuracy.

Figures 3–5 visualize the full log from complementary angles. Figure 3 exposes changes in MPJPE, speed, parameter count, and memory across all variants. Figure 4 shows the speed–error frontier with bubble area proportional to parameter count. Figure 5 normalizes the same four metrics, highlighting the M0 baseline, compact M6 draft model, and selected M9_RF2 model.

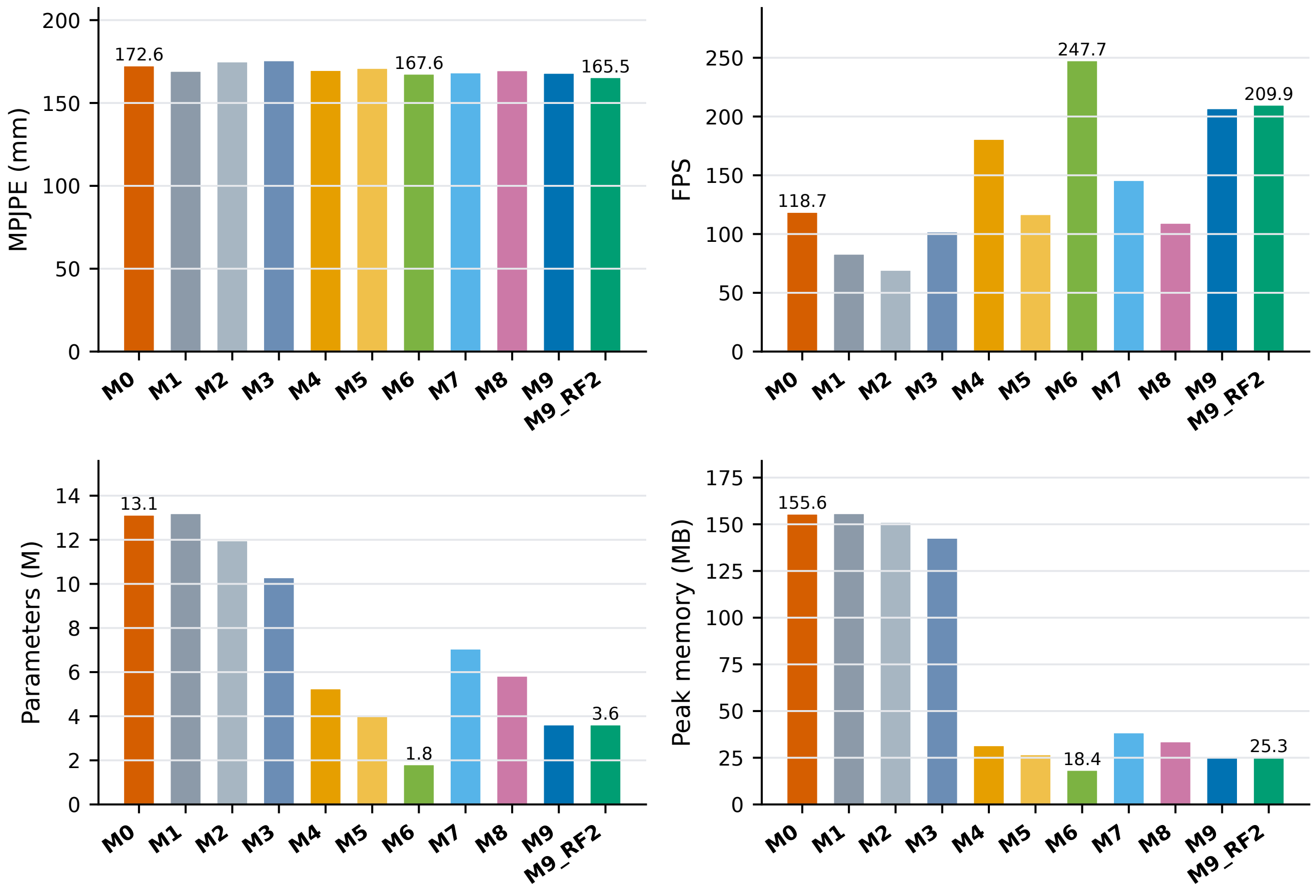

Fig. 3. Accuracy and efficiency across the full M0–M9 ablation and selected M9_RF2 evaluation.

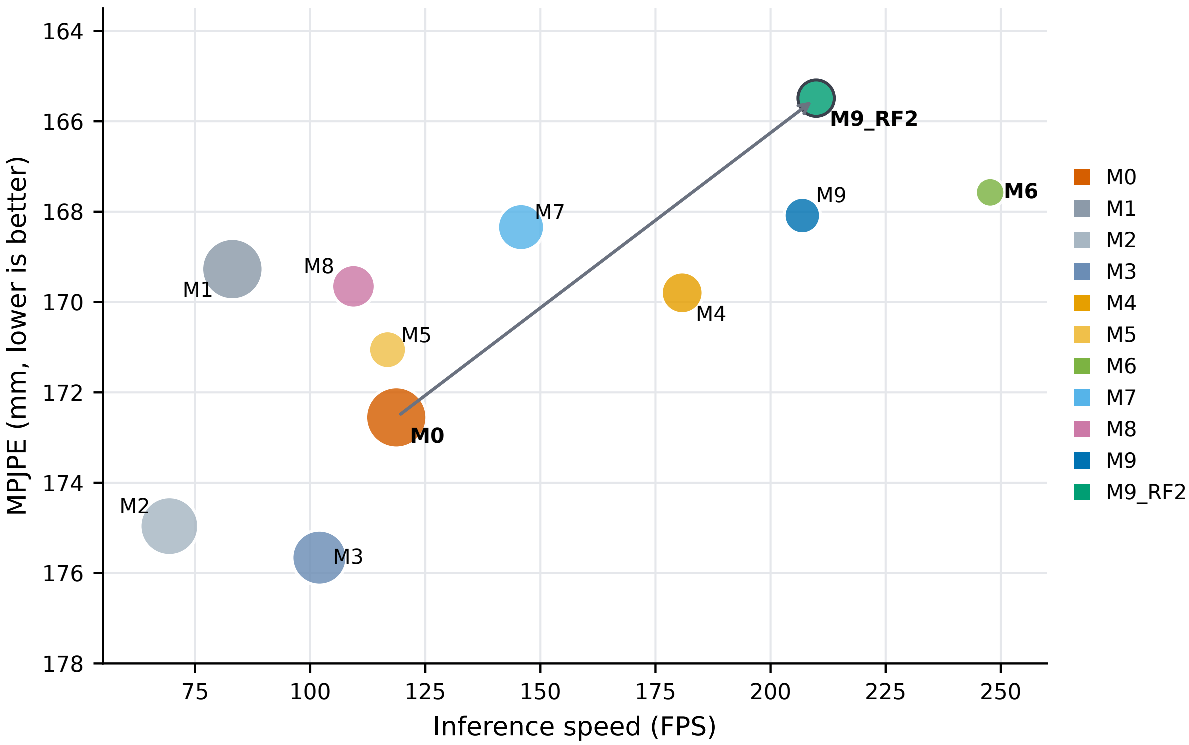

Fig. 4. Accuracy-efficiency trade-off of the evaluated variants. Bubble area corresponds to the number of parameters in millions.

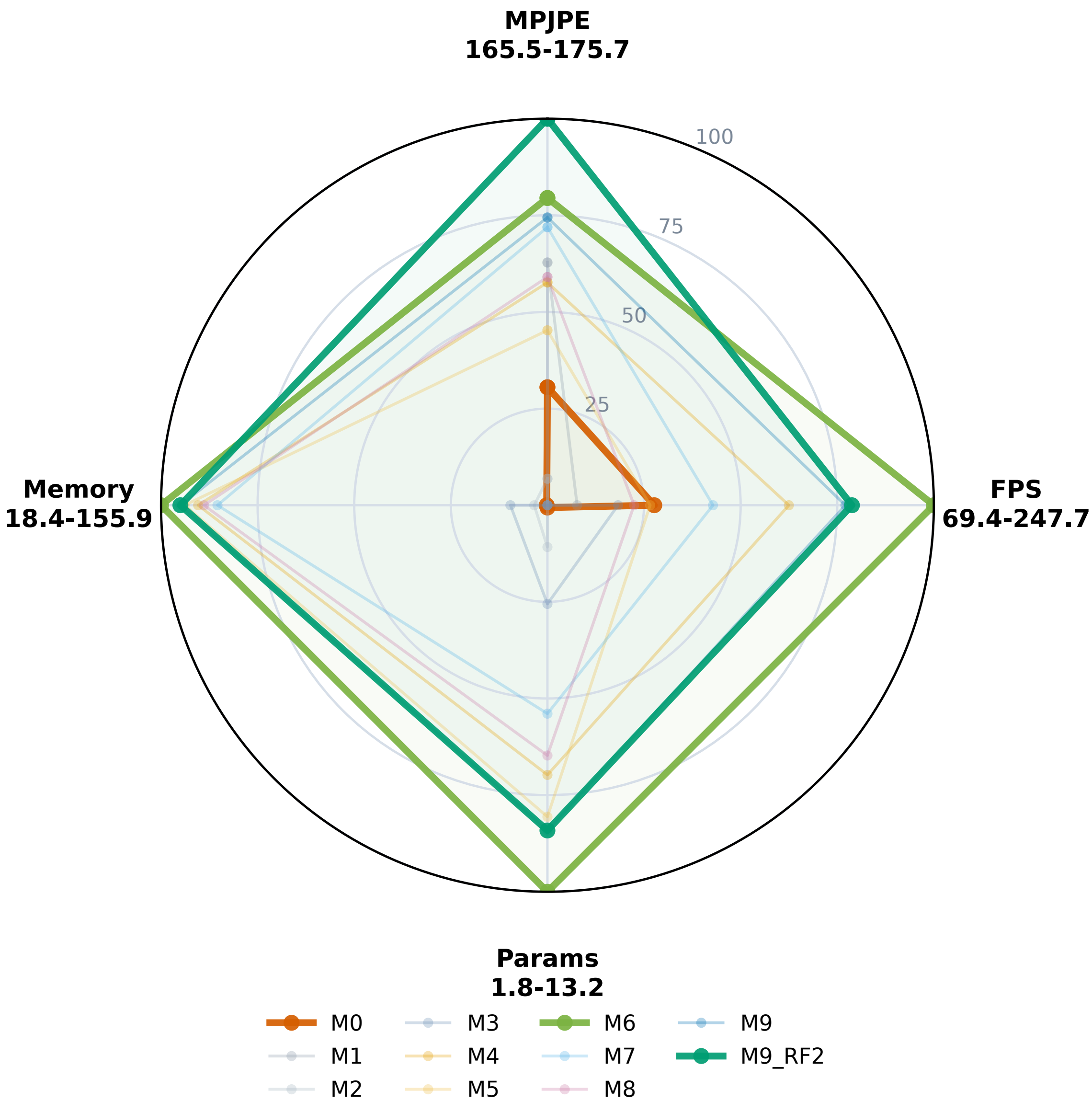

Fig. 5. Multi-metric comparison across the full ablation. Axes are min–max normalized so that a larger score is better; MPJPE, parameters, and memory are inverted before normalization.

B. Flow Solver Analysis

Table IV separates the draft-only model, disabled-flow inference, one-step flow, two-step flow, and retrained flow-loss variants. Figure 6 visualizes the two principal diagnostics for the same six configurations. The metrics expose a genuine trade-off: M9_RF2 attains the lowest overall MPJPE, whereas the draft-only M6 configuration produces the fewest unmatched persons. Among the tested M9 checkpoint settings, two-step Euler inference yields both the lowest MPJPE and the fewest unmatched persons.

TABLE IV
SUPPLEMENTARY FLOW ABLATION. LOWER MPJPE AND MATCHED-ONLY MPJPE ARE BETTER.

ID	Setting	MPJPE	Matched	Missed	1P	2P	3P
M6	Draft Mamba2-CSI	167.573	166.381	**54**	125.532	164.528	188.363
M9_no_flow	Flow disabled	167.732	165.919	82	122.870	163.026	191.421
M9	One-step flow	168.086	166.409	76	125.958	161.832	192.229
M9_RF2	Two-step flow	**165.487**	**163.886**	72	121.785	159.341	190.179
T_FW2_RF2	Retrained, two-step	167.981	166.082	86	121.335	160.970	194.804
T_FW2_RF4	Retrained, four-step	169.454	167.275	99	120.505	161.851	197.854

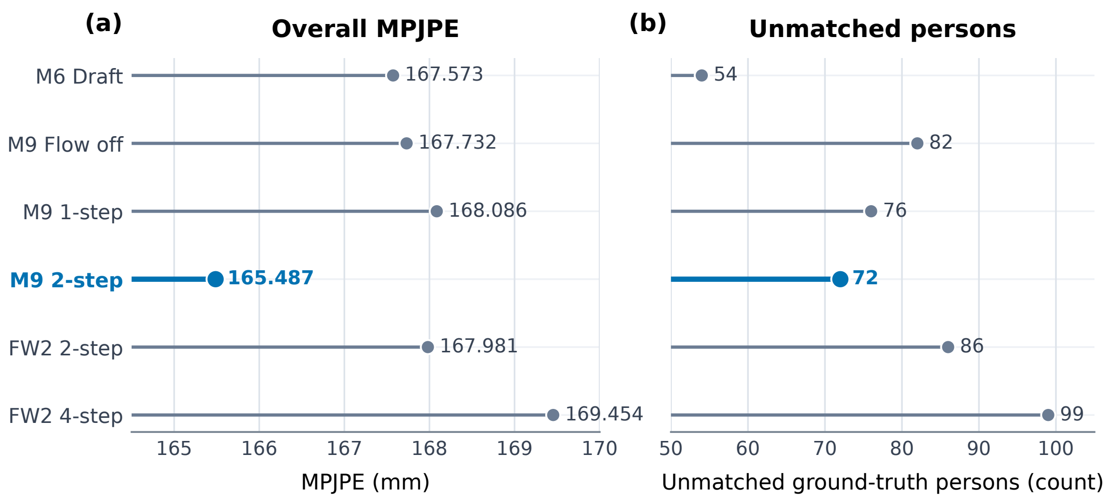

Fig. 6. Flow-solver analysis across the six evaluated configurations. Lower values are better in both panels: (a) overall MPJPE and (b) unmatched ground-truth person count. The highlighted point denotes the selected two-step M9 evaluation.

C. Per-Joint Error Analysis

Table V compares M9_RF2 with M0 at the joint level. Improvements concentrate on the left elbow, left wrist, hips, and knees, while distal right-side joints remain the hardest cases.

TABLE V
PER-JOINT MPJPE FOR THE SELECTED M9_RF2 MODEL VERSUS THE ORIGINAL PETR-STYLE M0 BASELINE, EVALUATED ON THE SAME TEST SPLIT AS TABLE III. LOWER IS BETTER. UNITS: MILLIMETERS.

joint	M9_RF2 (ours)	M0 (PETR)
neck	159.2	163.0
head	147.3	154.2
left shoulder	155.8	162.3
right shoulder	155.2	161.8
left elbow	126.9	139.3
right elbow	193.5	194.0
left wrist	128.7	138.1
right wrist	260.1	251.7
left hip	128.0	146.7
right hip	199.9	203.2
left knee	126.4	139.6
right knee	124.7	142.2
left ankle	132.9	141.6
right ankle	278.3	278.0
Mean	**165.5**	**172.6**

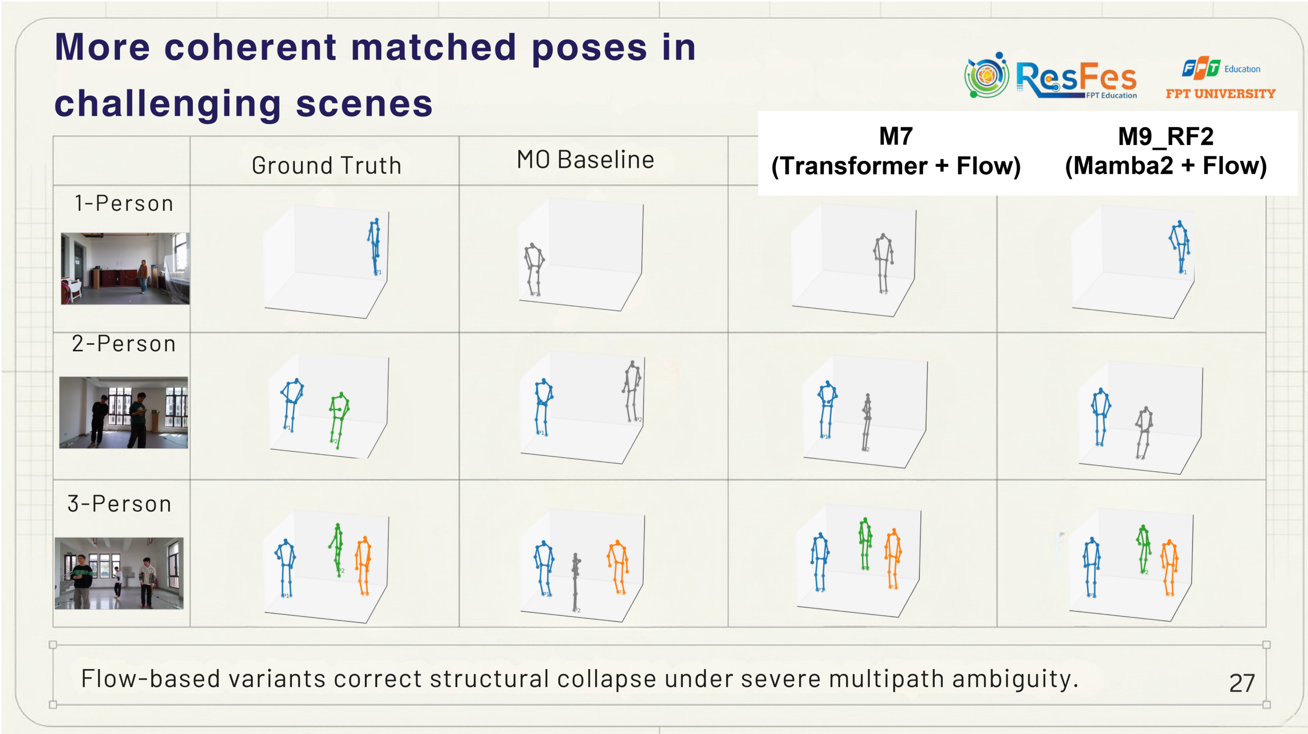

Fig. 7. Selected qualitative WiFi pose samples across 1-person, 2-person, and 3-person settings. Columns show the RGB scene, Kinect ground truth, M0, M7, and M9_RF2 predictions.

D. Receiver-Subset Diagnostic

Table VI reports the selected model under receiver masking. The full three-receiver input remains the intended fixed-layout operating condition; subset rows are included as a compact sensitivity diagnostic.

TABLE VI

PER-RECEIVER EVALUATION OF M9_RF2 ON THE SAME TEST SPLIT AS TABLE III. RECEIVERS NOT IN THE SUBSET ARE ZERO-MASKED AT THE INPUT OF THE WITIDAR PIPELINE (THE MODEL IS NOT RETRAINED). MIRRORS BASELINE TABLE 5 OF [1]. UNITS: MM; LOWER IS BETTER.

receiver subset	MPJPE	1P	2P	3P	matched	missed
3 receivers (R1+R2+R3)	165.5	121.8	159.3	190.2	15044	72
R1+R2	401.6	382.6	407.1	403.9	9567	5549
R1+R3	414.1	431.2	414.7	406.2	8838	6278
R2+R3	433.3	417.2	433.4	440.0	7828	7288
R2 only	498.0	496.3	499.6	497.0	498	14618

E. Failure Cases

Figure 7 shows selected qualitative samples across 1-person, 2-person, and 3-person settings. The examples compare the original M0 baseline with the M7 Transformer-flow model and selected M9_RF2 configuration; quantitative claims remain grounded in Tables III–V.

VII. DISCUSSION

The full ablation supports a narrow but useful conclusion: flow refinement is the small correction stage that lets the flattened Mamba2-CSI pipeline recover the best observed complete-model accuracy-efficiency point without returning to a heavier Transformer-style pose decoder. M6 establishes the efficiency of the compact draft path, M9 measures one-step refinement, and M9_RF2 supplies the selected two-step operating point. We accordingly read the title as a scoped architectural claim about this fixed-layout recipe, not as a claim that flow is sufficient in isolation.

The supplementary results qualify this conclusion without changing it. M6 attains fewer unmatched persons and higher throughput, whereas M9_RF2 improves overall and matched-only MPJPE. Per-joint results show that gains are not uniform across the skeleton, and receiver masking confirms that the reported operating point is tied to the intended three-receiver input.

VIII. LIMITATIONS

This study is limited to an internal fixed-layout ablation protocol and reports single-checkpoint results per configuration, so random-initialization variance is not quantified. All comparisons share the same controlled training budget, and the study follows the fixed-layout Person-in-WiFi 3D evaluation style described above. Confidence scores are still inherited from the draft stage, which can mis-rank refined poses when the flow update is large. The term flow refinement is used in a conditional correction sense; no reflow stage is performed. The spectral branch is best described as motion-aware rather than a direct Doppler estimator because it operates after feature projection. The privacy-sensitive framing refers to the absence of direct visual appearance capture and should not be read as a claim that CSI is identity-blind, since CSI-based person identification has been studied in prior work [38].

IX. CONCLUSION

This paper presents a compact draft-to-refine WiFi 3D pose pipeline. Under the internal fixed-layout ablation protocol, M9_RF2 achieves 165.487 mm MPJPE, improves over the re-evaluated M0 baseline by 7.067 mm, reaches 209.9 FPS, reduces parameters to 3.618M, and lowers peak memory to 25.29 MB. The full M0–M9 study and supplementary diagnostics support the subtitle directly: a flattened Mamba2-CSI backbone with lightweight two-step flow correction reaches the selected operating point without a complex pose-decoding architecture.

AUTHOR CONTRIBUTIONS

Nguyen Vinh Nghi, Le Ngoc Anh Thu, and Nguyen Ngo Nhat Nam contributed to system design, experiment execution, analysis, and manuscript preparation. Tran Ngoc Hoang supervised the research direction, reviewed the experimental interpretation, and advised the manuscript preparation.

AI DISCLOSURE

This manuscript was prepared with the assistance of AI-powered academic writing tools, in compliance with the ResFes 2026 commitment on responsible use of AI. AI assistance covered structure planning, result summarization, figure and table integration, and LaTeX formatting. All claims, numerical results, and conclusions were directed and reviewed by the authors, who take full responsibility for the accuracy and integrity of the work. No part of the experimental data, the model checkpoints, or the reported metrics was generated by an AI system.

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